

PQShield

HYBRID POST-QUANTUM SECURITY PLATFORM – TECHNICAL REPORT

AUTHOR	DOMAIN	TARGET ARCHITECTURE	STATUS
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Executive Summary: PQShield provides a software-based cybersecurity architecture designed to evaluate dual-layer hybrid post-quantum cryptography. By pairing classical algorithms with NIST-standardized quantum-resistant algorithms, the platform ensures data protection against both present-day eavesdropping and future quantum key-recoupling threats.

1. Research Context & Core Objectives

Quantum computing advances pose a existential risk to modern public-key cryptography (RSA, ECC), which relies on integer factorization and discrete logarithms. PQShield implements a hybrid migration defense, wrapping state-of-the-art classical primitives alongside post-quantum lattice-based primitives to secure telemetry from edge IoT nodes to cloud endpoints.

Dual-Layer Key Exchange (X25519 + ML-KEM)	ML-DSA-65 Quantum Digital Signature	AES-256-GCM Authenticated Payload Encryption
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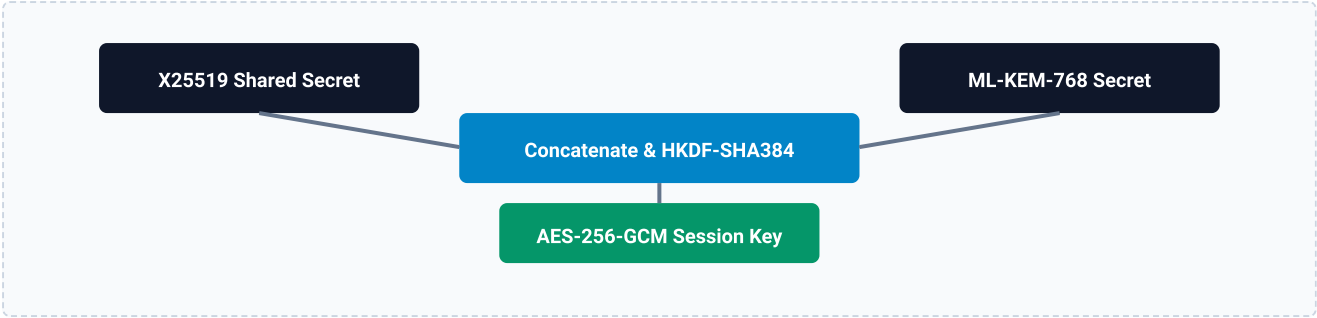
2. Cryptographic Primitive Suite

The platform standardizes on a precise suite of classical and post-quantum cryptographic primitives to maintain high security margins without compromising IoT throughput:

Security Function	Algorithm Implemented	Standardization / Spec
Classical Key Agreement	X25519 (ECDH over Curve25519)	RFC 7748
Post-Quantum KEM	ML-KEM-768 (CRYSTALS-Kyber)	NIST FIPS 203
Key Derivation Function	HKDF-SHA384	RFC 5869
Payload Encryption	AES-256-GCM	NIST SP 800-38D
Digital Signatures	ML-DSA-65 (CRYSTALS-Dilithium)	NIST FIPS 204

3. Hybrid Cryptographic Construction

To prevent single-point failures in either classical or newly deployed post-quantum schemes, key agreement performs a dual derivation. Shared secrets from both X25519 and ML-KEM-768 are concatenated prior to HKDF ingestion.



4. End-to-End System Architecture

The platform encapsulates simulated IoT edge nodes, an MQTT message broker, a Python/FastAPI secure gateway, and a modern React monitoring dashboard backed by PostgreSQL.

Component	Technology	Role in Platform
Virtual Devices	Python / Simulator Engine	Generates encrypted telemetry payloads with sequence counters.
Message Broker	Eclipse Mosquitto (MQTT)	Transports encrypted binary packages across topic queues.
Secure Gateway	FastAPI / SQLAlchemy	Verifies ML-DSA-65 signatures, checks replay windows, decrypts data.
Audit Storage	PostgreSQL	Persists security decisions, anomaly events, and benchmark results.
Security UI	React / TypeScript / Tailwind	Provides real-time inspection, key lab, and metric visualization.

5. IoT Telemetry Protection & Replay Prevention

Telemetry payload packages are formatted as canonical binary blobs containing device identifier, sequence counter, ciphertext, and ML-DSA-65 signature.

- Sequence-Based Replay Protection:** The gateway tracks monotonic sequence numbers per device. Out-of-order or duplicate sequences trigger immediate drop events.
- Strict Topic Scoping:** Devices are restricted to publish on `pqshield/<device_id>/telemetry` . Misaligned IDs result in security alerts.

6. Cryptographic Performance Benchmarking

Comparative analysis was executed across Classical, Post-Quantum (PQC), and Hybrid configurations to measure computational latency and transport overhead.

Metric / Step	Classical (X25519+Ed25519)	PQC Only (ML-KEM+ML-DSA)	Hybrid Construction
Key Generation	~0.12 ms	~0.85 ms	~0.98 ms
Encapsulation / Key Agreement	~0.15 ms	~0.42 ms	~0.58 ms
Signature Generation	~0.08 ms	~1.25 ms	~1.27 ms
Signature Verification	~0.22 ms	~0.65 ms	~0.68 ms
Public Key Size	32 + 32 B	1,184 + 1,952 B	1,216 + 1,984 B
Package Overhead	~128 Bytes	~3,800 Bytes	~3,930 Bytes

7. Scalability & Load Testing

Integrating Locust load tests demonstrated stable gateway throughput up to 500 concurrent virtual device connections under steady-state simulated MQTT telemetry generation, maintaining an average API processing latency under 18ms.

8. Limitations & Future Scope

- Key Storage Memory Scope:** Ephemeral private key material is maintained in process memory. Future production builds will integrate HashiCorp Vault / Cloud KMS.
- In-Memory Replay Cache:** Replay windows currently reset upon gateway process restart; planned migration will utilize Redis.
- Hardware Validation:** Migration of simulated edge nodes to physical microcontroller hardware (ESP32-S3 / Raspberry Pi Pico 2).

9. Conclusion

PQShield proves the feasibility of deploying hybrid post-quantum security layers on modern IoT-to-Cloud architectures. While post-quantum algorithms increase public key and signature size, the hybrid key agreement guarantees that confidentiality remains intact even if one cryptographic primitive is broken in the future.

